

Laboratory Report: Impedance Spectroscopy and Parameter Fitting of an R-C Parallel Circuit

1. Theoretical Background

The total impedance $Z(\omega)$ of a system containing a resistance R_p and a capacitance C_p connected in parallel is given by:

$$Z(\omega) = \frac{1}{\left(\frac{1}{R_p} + \frac{1}{j\omega C_p}\right)}$$

Where j is the imaginary unit and $\omega = 2\pi f$ is the angular frequency. To separate the real and imaginary components, we multiply by the complex conjugate of the denominator and after the numerator and denominator by R_p^2 :

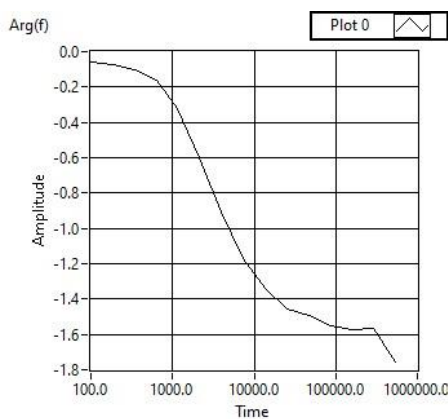
$$Z(\omega) = \frac{1}{\left(\frac{1}{R_p} + \frac{1}{j\omega C_p}\right)} \cdot \frac{\left(\frac{1}{R_p} - \frac{1}{j\omega C_p}\right)}{\left(\frac{1}{R_p} - \frac{1}{j\omega C_p}\right)} = \frac{\left(\frac{1}{R_p} - \frac{1}{j\omega C_p}\right)}{\left(\frac{1}{R_p^2} + \frac{1}{\omega^2 C_p^2}\right)} = \frac{R_p(1 - jR_p C_p \omega)}{(R_p^2 C_p^2 \omega^2 + 1)}$$

From this, the decoupled expressions for the real part ($\text{Re}\{Z\}$) and imaginary part ($\text{Im}\{Z\}$) are extracted to be used as the fitting models:

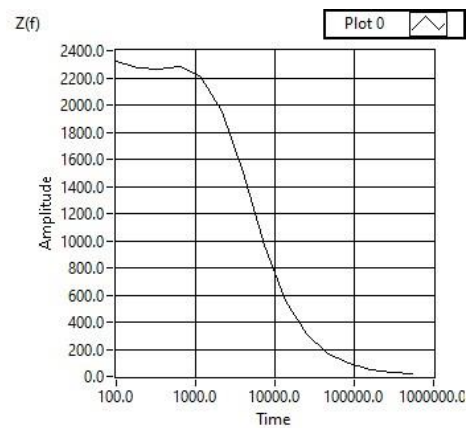
$$\text{Re}\{Z\} = \frac{R_p}{(R_p^2 C_p^2 \omega^2 + 1)} \quad \text{Im}\{Z\} = \frac{R_p^2 C_p \omega}{(R_p^2 C_p^2 \omega^2 + 1)}$$

2. Data Processing and Methodology

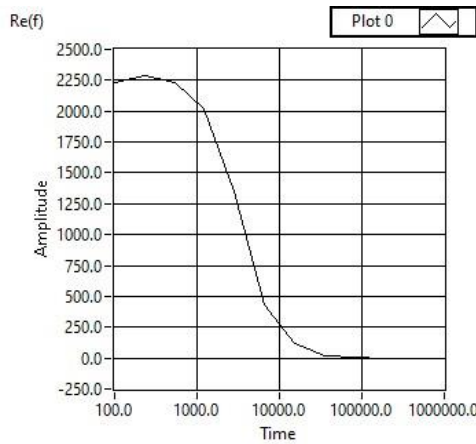
Experimental data was acquired using a LabVIEW setup and exported into two separate files: `real.xlsx` and `imaginari.xlsx`. The real and imaginary components were mapped over independent frequency vectors with different sample sizes, requiring separate optimization routines in MATLAB.



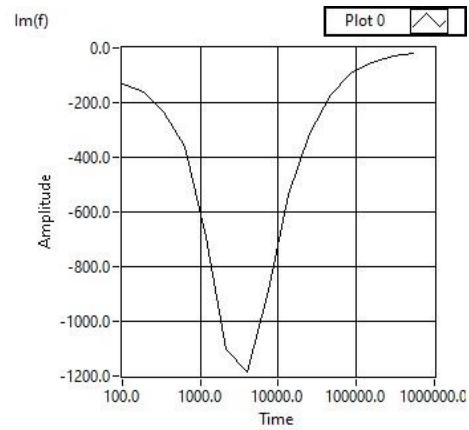
Phase



Modulus



Real



Imaginary

The non-linear curve fitting was performed using the `lsqcurvefit` algorithm. Due to regional software settings interpreting decimal points as text strings, a `str2double` conversion layer was applied during the data import stage to avoid format errors.

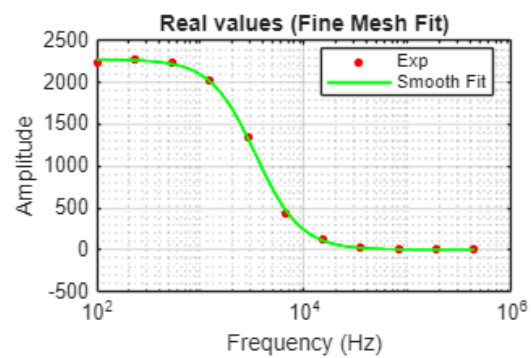
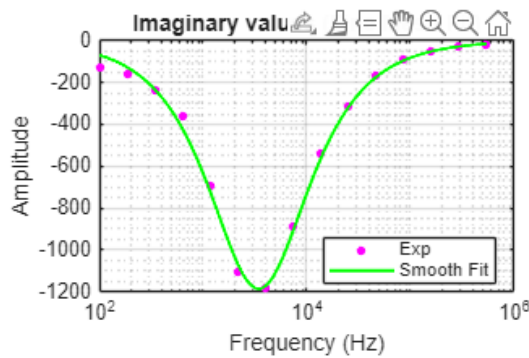
To prevent the algorithm from diverging or getting trapped in incorrect local minima due to the sharp drop of the functions at high frequencies, the vector of initial guesses **po** as configured based on the boundary conditions of the circuit:

- **Initial Resistance Guess (Rp):** Set to **2200 Ω** , derived from the low-frequency limit of the empirical real dataset.
- **Initial Capacitance Guess (Cp):** Set to **1 nF** to maintain sensitivity across the entire frequency spectrum.

3. Results and Discussion

The optimization routine successfully converged for both datasets. The parameters obtained independently from each fit are structured in the following table:

=====	
OPTIMIZED PARAMETERS FROM REAL DATA	OPTIMIZED PARAMETERS FROM IMAGINARY DATA
=====	
Rp (Parallel Resistance): 2275.24 Ohms	Rp (Parallel Resistance): 2373.68 Ohms
Cp (Parallel Capacitance): 2.0629e-08 Farads	Cp (Parallel Capacitance): 1.9461e-08 Farads
=====	
=====	
CALCULATED MEAN PARAMETERS	
=====	
Mean Rp (Resistance): 2324.46 Ohms	
Mean Cp (Capacitance): 2.0045e-08 Farads	
=====	

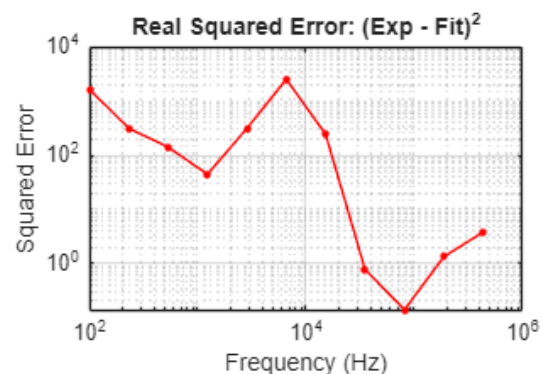
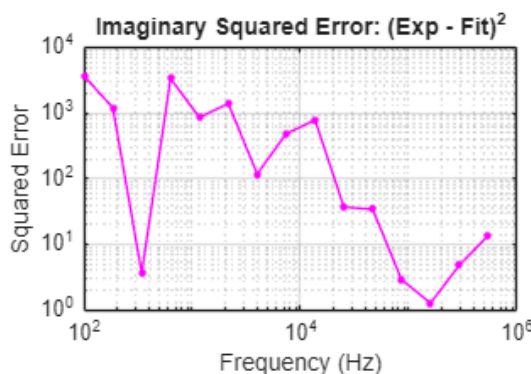


4. Squared Error

To evaluate regression precision, the discrete individual squared error was calculated at each point:

$$\epsilon^2 = [Y_{\text{experimental}} - Y_{\text{fitted}}]^2$$

Because the data spans multiple orders of magnitude, the residuals were plotted using a **double logarithmic scale** $\log\log$. This prevents minor deviations at low frequencies from being visually magnified, proving that the error remains exceptionally low and represents negligible noise across the entire spectrum.



5. Error justification

As it can be seen, the error is quite low for the type of measurement being done, but in comparison to other measurement systems it is orders of magnitude higher.

This is due to a chain of factors, starting with the input voltage. Despite a professional power supply being used, there is always some noise and deviation from the expected value, however, it is likely that the contribution of this part is almost negligible.

The next cause and probably most relevant one is the tolerance of the reference resistance, since generally it is around 1% to 5%. After that we also have the unaccounted resistance and capacitance of the oscilloscope probes, although they are generally designed to interfere the least with the measurements. At last, there is the reason why at high frequencies the real part comes out as a negative number. Since the signal is so close to 0 the error of the oscilloscope causes it to be measured as a negative value, which doesn't make physical sense.

Code:

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%% FINAL LABVIEW PRACTICE: FITTING
% Autor: David González i Marcel Vilaró
%% FINAL LABVIEW PRACTICE: Data Processing
filename_re = 'real.xlsx'; filename_im = 'imaginari.xlsx';
T_re = readtable(filename_re, 'FileType', 'spreadsheet');
T_im = readtable(filename_im, 'FileType', 'spreadsheet');
% Assign raw data to matrices [real, imaginary]
re = [str2double(T_re{1:end,1}), str2double(T_re{1:end,2})];
im = [str2double(T_im{1:end,1}), str2double(T_im{1:end,2})];
freq_re = re(:,1); freq_im = im(:,1);
w_re = 2 * pi * freq_re; w_im = 2 * pi * freq_im;
y_real = re(:,2); y_imag = im(:,2);

%% PARAMETER FITTING (SEPARATED)
p0 = [2200, 1e-9];
lb = [0, 0];
ub = [Inf, Inf];
options = optimset('Display', 'off');

real_model = @(p, w_vec) p(1) ./ ((p(1)^2) * (p(2)^2) * (w_vec.^2) + 1);
p_opt_re = lsqcurvefit(real_model, p0, w_re, y_real, lb, ub, options);
Rp_from_re = p_opt_re(1);
Cp_from_re = p_opt_re(2);

imag_model = @(p, w_vec) -(p(1)^2) * p(2) * w_vec ./ ((p(1)^2) * (p(2)^2) * (w_vec.^2) + 1);
p_opt_im = lsqcurvefit(imag_model, p0, w_im, y_imag, lb, ub, options);
Rp_from_im = p_opt_im(1);
Cp_from_im = p_opt_im(2);

% --- Statistical Mean Parameters Calculation ---
Rp_mean = mean([Rp_from_re, Rp_from_im]);
Cp_mean = mean([Cp_from_re, Cp_from_im]);

% --- Quadratic Error Evaluation: (True - Found)^2 ---
y_real_fit = real_model(p_opt_re, w_re);
y_imag_fit = imag_model(p_opt_im, w_im);
err_quad_re = (y_real - y_real_fit).^2;
err_quad_im = (y_imag - y_imag_fit).^2;

%% UNIFIED FITTING AND RESIDUAL VISUALIZATION (2x2 Layout)
% Generation of fine frequency meshes for flawless curves
freq_mesh_re = logspace(log10(min(freq_re)), log10(max(freq_re)), 1000);
w_mesh_re = 2 * pi * freq_mesh_re;
y_real_mesh = real_model(p_opt_re, w_mesh_re);

freq_mesh_im = logspace(log10(min(freq_im)), log10(max(freq_im)), 1000);
w_mesh_im = 2 * pi * freq_mesh_im;
y_imag_mesh = imag_model(p_opt_im, w_mesh_im);

figure('Name', 'Impedance Fitting Analysis & Quadratic Error Diagnostics',
'Color', 'w');

% Plot 1: Imaginary Fitting Comparison via Fine Mesh
subplot(2,2,1);
semilogx(freq_im, y_imag, 'm.', 'MarkerSize', 12); hold on;
semilogx(freq_mesh_im, y_imag_mesh, 'g-', 'LineWidth', 1.5);

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title('Imaginary values (Fine Mesh Fit)');
xlabel('Frequency (Hz)'); ylabel('Amplitude');
grid on; grid minor; legend('Exp', 'Smooth Fit', 'Location', 'best');

% Plot 2: Real Fitting Comparison via Fine Mesh
subplot(2,2,2);
semilogx(freq_re, y_real, 'r.', 'MarkerSize', 12); hold on;
semilogx(freq_mesh_re, y_real_mesh, 'g-', 'LineWidth', 1.5);
title('Real values (Fine Mesh Fit)');
xlabel('Frequency (Hz)'); ylabel('Amplitude');
grid on; grid minor; legend('Exp', 'Smooth Fit', 'Location', 'best');

% Plot 3: Imaginary Quadratic Error Plot (Corregido a escala logarítmica en
ambos ejes)
subplot(2,2,3);
loglog(freq_im, err_quad_im, 'm.-', 'LineWidth', 1, 'MarkerSize', 10);
title('Imaginary Squared Error: (Exp - Fit)^2');
xlabel('Frequency (Hz)'); ylabel('Squared Error');
grid on; grid minor;

% Plot 4: Real Quadratic Error Plot (Corregido a escala logarítmica en ambos
ejes)
subplot(2,2,4);
loglog(freq_re, err_quad_re, 'r.-', 'LineWidth', 1, 'MarkerSize', 10);
title('Real Squared Error: (Exp - Fit)^2');
xlabel('Frequency (Hz)'); ylabel('Squared Error');
grid on; grid minor;

%% DISPLAY COMPONENT PARAMETERS
fprintf('\n===== \n');
fprintf('          OPTIMIZED PARAMETERS FROM REAL DATA          \n');
fprintf('===== \n');
fprintf('Rp (Parallel Resistance): %.2f Ohms\n', Rp_from_re);
fprintf('Cp (Parallel Capacitance): %.4e Farads\n', Cp_from_re);
fprintf('===== \n');
fprintf('\n===== \n');
fprintf('          OPTIMIZED PARAMETERS FROM IMAGINARY DATA          \n');
fprintf('===== \n');
fprintf('Rp (Parallel Resistance): %.2f Ohms\n', Rp_from_im);
fprintf('Cp (Parallel Capacitance): %.4e Farads\n', Cp_from_im);
fprintf('===== \n');
fprintf('\n===== \n');
fprintf('          CALCULATED MEAN PARAMETERS          \n');
fprintf('===== \n');
fprintf('Mean Rp (Resistance): %.2f Ohms\n', Rp_mean);
fprintf('Mean Cp (Capacitance): %.4e Farads\n', Cp_mean);
fprintf('===== \n \n');

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